

ABSTRACT

By orthogonalizing the primary inertia moment vector produced by an engine with many eigen vectors, a new engine mounting method called "the orthogonal engine mount system" was designed.

This paper explains its theoretical background. In addition, a motorcycle with a two-cylinder engine incorporating this system was analyzed by means of modal analysis and building block approach techniques to determine the effectiveness of the orthogonal engine mount system.

MOST OF THE VIBRATIONAL PROBLEMS OF MOTORCYCLES are caused by the unbalanced inertia forces and/or moments generated by the engine.

In the past, the control of the amplitude and direction of these forces has been carried out by changing the balance of crankwebs (1)*, but because the demand for low vibration has increased, effort has been exerted to minimize the unbalanced forces by the method of installing balancer shafts, making multi-cylinder engines and so on.

Also, the practice of mounting engines in the chassis with vibration-isolating support has been adopted. But new problems have arisen in chain driven motorcycles, such as the jumping of engines due to the reaction of the rear wheel drive, and the lowering of stability and handling characteristics due to the reduction of frame stiffness.

Consequently, the original goal of vibration isolation has often been sacrificed for heightening the stiffness of engine support. Accordingly, one of the solutions is to take the direction of the force generated by the engine properly and

*Numbers in parentheses designate references at end of paper.

generate vibration only in one direction, take the resonant rotating speed in that direction below the usable operating speed range and make the supporting stiffness in other directions high as the situation demands.

However, in engines mounted on motorcycles, generally the principal axes of inertia are inclined largely owing to their configuration, and dynamic coupling due to the products of inertia occur. Even if the direction of force is selected properly, the direction of vibration does not necessarily coincide with this direction, and therefore, fairly complicated vibration phenomena occur.

direction
of forces
≠
direction
of vibration

In this paper, the theory of the orthogonal engine mount system and its application to motorcycles is described. The static coupling due to the stiffness of the engine support is positively utilized against the dynamic coupling. The vibration-isolating function and the stiff supporting function are compatible when the primary inertia moment generated by two-cylinder or three-cylinder engine is the main exciting source.

THEORY

PREPARATION—The linear equation of motion of n-degree of freedom undamped system in matrix form, expressed in generalize coordinates $q_1, q_2, q_3, \dots, q_n$ can be written as (2):

$$[M][\ddot{q}] + [K][q] = [f] \quad (1)$$

Where $[M]$: mass matrix
 $[K]$: stiffness matrix
 $[f]$: external force vector

In order to determine the steady-state frequency response in Equation (1), assume that $[q] = [Q]e^{i\omega t}$ and $[f] = [F]e^{i\omega t}$, and substitute these equations into Equation (1),

$$\{-\omega^2 [M] + [K]\} [Q] = [F] \quad (2)$$

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$$M \ddot{q} + K q = f$$

$$\text{with } q = Q e^{j\omega t}$$

$$f = F e^{j\omega t}$$

$$[-\omega^2 M + K] Q = F$$

too soft mounting
results in stability
problems and
chain jumping

$$[M][\ddot{q}] + [K][q] = [f]$$

To determine the eigenvalues (natural frequencies) ω_r and the eigenvectors (normal modes) $[u^r]$ ($r = 1 - n$), substitute $[F] = [0]$ into equation (2),

$$\{-\omega^2[M] + [K]\}[Q] = [0] \quad (3)$$

These eigenvectors have an important property called orthogonal conditions. Namely,

$$[u^s]^T [M] [u^r] = \begin{cases} 0 & (s \neq r) \\ m_r & (s = r) \end{cases} \quad (4)$$

$$[u^s]^T [K] [u^r] = \begin{cases} 0 & (s \neq r) \\ k_r & (s = r) \end{cases} \quad (5)$$

Expressing the solution of Equation (2) with the linear combination of the eigenvectors, and using the orthogonal conditions described above, the steady-state frequency response in Equation (1) is expressed as the following equation.

$$[Q] = \sum_{r=1}^n \frac{[u^r]^T [F] [u^r]}{m_r (-\omega^2 + \omega_r^2)} \quad (\omega_r^2 = \frac{k_r}{m_r}) \quad (6)$$

Here, as shown in Figure 1, the center of gravity of the engine is taken as the origin of the coordinate system with the x-axis in the longitudinal direction, the y-axis in the lateral direction and the z-axis in the vertical direction. The following assumptions are made.

- The engine is a rigid body.
- The translational displacements and rotational displacements of the engine are small.
- Each resilient supporting element is a linear spring which does not have mass or damping.
- The chassis is a rigid body with infinite mass (ground).
- One of the principal axes of inertia of the engine coincides with y-axis.
- The center of elasticity of all supporting elements coincides with the center of gravity.
- One of the principal axes of elasticity of all supporting elements coincides with y-axis.
- The only external force is the unbalanced primary inertia moment around x-axis.

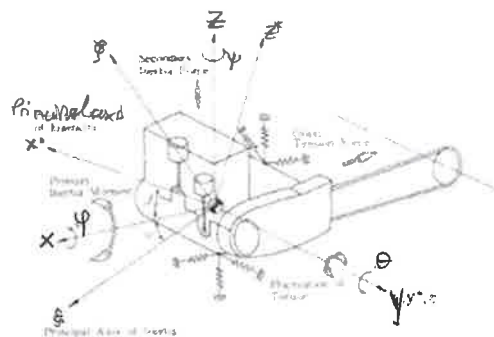
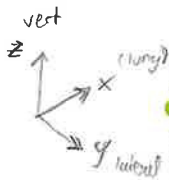


Fig. 1 Coordinate System

Applying these assumptions to Equation (1), $n = 6$ and we have

$$\begin{bmatrix} m & 0 & 0 & 0 & 0 & 0 \\ 0 & m & 0 & 0 & 0 & 0 \\ 0 & 0 & m & 0 & 0 & 0 \\ 0 & 0 & 0 & I_x & 0 & -I_{zx} \\ 0 & 0 & 0 & 0 & I_y & 0 \\ 0 & 0 & 0 & -I_{zx} & 0 & I_z \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ \phi \\ \theta \\ \psi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$+ \begin{bmatrix} K_{xx} & 0 & K_{xz} & 0 & 0 & 0 \\ 0 & K_{yy} & 0 & 0 & 0 & 0 \\ K_{xz} & 0 & K_{zz} & 0 & 0 & 0 \\ 0 & 0 & 0 & K_{\phi\phi} & 0 & K_{\phi\psi} \\ 0 & 0 & 0 & 0 & K_{\theta\theta} & 0 \\ 0 & 0 & 0 & K_{\phi\psi} & 0 & K_{\psi\psi} \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ \phi \\ \theta \\ \psi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ f_\phi \\ 0 \\ 0 \end{bmatrix}$$

Where m : mass of the engine

I : moment or product of inertia of the engine (subscripts show the direction)

K : stiffness coefficient of supporting elements (subscripts show the direction)

x_c : displacement of center of gravity in x-direction

y_c : displacement of center of gravity in y-direction

z_c : displacement of center of gravity in z-direction

ϕ : rotational displacement around x-axis

θ : rotational displacement around y-axis

ψ : rotational displacement around z-axis

f_ϕ : exciting moment around x-axis

The eigenvectors take the form of the following equation.

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ u_\phi^1 \\ 0 \\ u_\psi^1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ u_\phi^2 \\ 0 \\ u_\psi^2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} u_x^3 \\ 0 \\ u_z^3 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} u_x^4 \\ u_y^4 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (7)$$

Eigenvectors

Steady-state frequency response is obtained from Equation (6).

Steady State

freq. response

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ \phi \\ \theta \\ \psi \end{bmatrix} = \sum_{r=1}^6 \frac{[u^r]^T [F] [u^r]}{m_r (-\omega^2 + \omega_r^2)} \begin{bmatrix} 0 \\ 0 \\ 0 \\ f_\phi \\ 0 \\ 0 \end{bmatrix} \quad (8)$$

Substituting Equation (7) into Equation (8),

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ \phi \\ \theta \\ \psi \end{bmatrix} = \sum_{r=1}^6 \frac{u_r^T F_\phi}{m_r (-\omega^2 + \omega_r^2)} \begin{bmatrix} 0 \\ 0 \\ 0 \\ u_\phi^r \\ 0 \\ u_\psi^r \end{bmatrix} \quad (9)$$

From the above assumptions, the vibrations generated turn out to be only those around x-axis and around z-axis, and it is known that two resonances are presented.

From Equation (3) we have

$$\left\{ -\omega^2 \begin{bmatrix} I_x - I_{zx} & 0 \\ -I_{zx} & I_z \end{bmatrix} + \begin{bmatrix} K_{\phi\phi} & K_{\phi\psi} \\ K_{\psi\phi} & K_{\psi\psi} \end{bmatrix} \right\} \begin{bmatrix} \phi \\ \psi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Namely, solving

$$\begin{bmatrix} K_{\phi\phi} - \omega^2 I_x & K_{\phi\psi} + \omega^2 I_{zx} \\ K_{\psi\phi} + \omega^2 I_{zx} & K_{\psi\psi} - \omega^2 I_z \end{bmatrix} \begin{bmatrix} \phi \\ \psi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (10)$$

$\omega_1, \omega_2, [u_\phi^1 \ u_\psi^1]^T, [u_\phi^2 \ u_\psi^2]^T$ are obtained. Moreover, if

$$[u_\phi^2 \ u_\psi^2]^T = [0 \ 1]^T \quad (11)$$

can be assumed, substituting Equation (11) into Equation (9) gives

$$\begin{bmatrix} X_C \\ Y_C \\ Z_C \\ \phi \\ \psi \end{bmatrix} = \frac{u_\phi^1 F_\phi}{m_1 (-\omega^2 + \omega_1^2)} \begin{bmatrix} 0 \\ 0 \\ 0 \\ u_\phi^1 \\ u_\psi^1 \end{bmatrix} \quad (12)$$

Namely, the system has only one resonant frequency ω_1 , and the axis of rotational vibration is fixed regardless of frequency. This can be expressed as follows. The exciting force vector is in the orthogonal condition with the vectors of other modes than the first mode. By this, the design calculation with good prospect can be performed.

DERIVATION OF EQUATIONS OF CONDITIONS-
The conditions under which Equation (1) holds when $\omega_1 \neq \omega_2$,

$$[u_\phi^1 \ u_\psi^1] \begin{bmatrix} I_x & -I_{zx} \\ -I_{zx} & I_z \end{bmatrix} \begin{bmatrix} u_\phi^2 \\ u_\psi^2 \end{bmatrix} = 0 \quad (13)$$

$$[u_\phi^1 \ u_\psi^1] \begin{bmatrix} K_{\phi\phi} & K_{\phi\psi} \\ K_{\psi\phi} & K_{\psi\psi} \end{bmatrix} \begin{bmatrix} u_\phi^2 \\ u_\psi^2 \end{bmatrix} = 0 \quad (14)$$

Substituting Equation (11) into Equation (13),

$$-I_{zx} u_\phi^1 + I_z u_\psi^1 = 0$$

Substituting Equation (11) into Equation (14),

$$K_{\phi\psi} u_\phi^1 + K_{\psi\psi} u_\psi^1 = 0$$

To have the solution in which both u_ϕ^1 and u_ψ^1 are not zero,

$$\begin{vmatrix} -I_{zx} & I_z \\ K_{\phi\psi} & K_{\psi\psi} \end{vmatrix} = 0$$

Therefore,

$$I_z K_{\phi\psi} + I_{zx} K_{\psi\psi} = 0 \quad (15)$$

is required. Rewriting Equation (15),

$$\frac{I_{zx}}{I_z} = -\frac{K_{\phi\psi}}{K_{\psi\psi}} = R \quad (16)$$

On the contrary, substituting Equation (16) into Equation (10), and eliminating I_{zx} and $K_{\phi\psi}$,

$$\begin{bmatrix} K_{\phi\phi} - \omega^2 I_x & -RK_{\psi\psi} + \omega^2 I_z \\ -RK_{\psi\psi} + \omega^2 I_z & K_{\psi\psi} - \omega^2 I_z \end{bmatrix} \begin{bmatrix} \phi \\ \psi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (17)$$

Let the determinant of the coefficient matrix on the left side of Equation (17) be zero,

$$\{K_{\phi\phi} - R^2 K_{\psi\psi} - \omega^2 (I_x - R^2 I_z)\} (K_{\psi\psi} - \omega^2 I_z) = 0$$

Therefore,

$$\omega_1^2 = \frac{K_{\phi\phi} - R^2 K_{\psi\psi}}{I_x - R^2 I_z}, \quad \omega_2^2 = \frac{K_{\psi\psi}}{I_z} \quad (18)$$

When $\omega = \omega_2$, Equation (17) turns out to be

$$\begin{bmatrix} K_{\phi\phi} - \frac{K_{\psi\psi} I_x}{I_z} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \phi \\ \psi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (19)$$

Here, if $K_{\phi\phi}/K_{\psi\psi} = I_x/I_z$ holds, substituting this into Equation (18)

$$\omega_1^2 = \frac{K_{\psi\psi}}{I_z} = \omega_2^2$$

results in, and this is contrary to the premise $\omega_1 \neq \omega_2$. Accordingly, from $K_{\phi\phi}/K_{\psi\psi} \neq I_x/I_z$, Equation (19) is solved.

$$[u_\phi^2 \ u_\psi^2]^T = [0 \ 1]^T$$

According to the above description, the necessary and sufficient condition under which Equation (11) holds when $\omega_1 \neq \omega_2$, is Equation (16).

Next, as shown in Figure 1, denoting the inclination of the principal axis of inertia by α (positive rotation about y-axis is one which advances a right-handed screw in the positive direction of the axis), and the principal moments of inertia by I_ξ and I_ζ ,

$$\begin{aligned} I_x &= I_\xi \cos^2 \alpha + I_\zeta \sin^2 \alpha \\ I_z &= I_\xi \sin^2 \alpha + I_\zeta \cos^2 \alpha \\ I_{zx} &= (I_\xi - I_\zeta) \sin \alpha \cos \alpha \end{aligned} \quad (20)$$

Similarly, denoting the inclination of the principal axis of elasticity by β and the principal restoring coefficient by K_ϕ^* and K_ψ^* ,

$$\begin{aligned} K_{\phi\phi} &= K_\phi^* \cos^2 \beta + K_\psi^* \sin^2 \beta \\ K_{\psi\psi} &= K_\phi^* \sin^2 \beta + K_\psi^* \cos^2 \beta \\ K_{\phi\psi} &= -(K_\phi^* - K_\psi^*) \sin \beta \cos \beta \end{aligned} \quad (21)$$

Substituting Equations (20) and (21) into Equation (16),

in this case,
SDOF condition
(→ decoupling)

this holds if;

$$\frac{(I_x - I_z) \sin \alpha \cos \alpha}{I_x \sin^2 \alpha + I_z \cos^2 \alpha} = \frac{(K_x^* - K_y^*) \sin \beta \cos \beta}{K_x^* \sin^2 \beta + K_y^* \cos^2 \beta} = R$$

Expressing as $I_x/I_z = p$ and $K_x^*/K_y^* = q$,

$$\frac{(p-1) \tan \alpha}{p \tan^2 \alpha + 1} = \frac{(q-1) \tan \beta}{q \tan^2 \beta + 1} = R \quad (22)$$

According to Equation (22), the relation between q and β using R as a parameter is shown in Figure 2. Replacing p for q and α for β respectively, it also represents the relation between p and α .

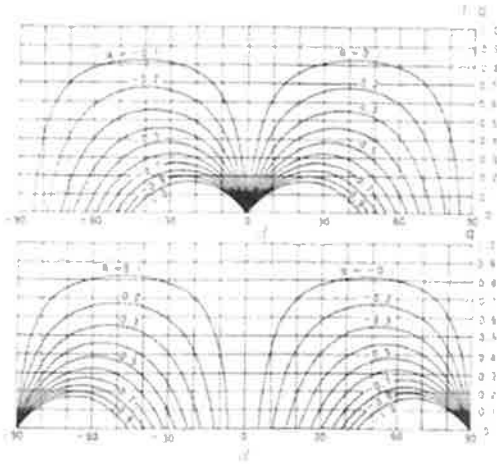


Fig. 2 Relationship between q and β against R as a Parameter

STEADY-STATE FREQUENCY RESPONSE AND RESONANT FREQUENCY-Again, substituting Equations (11) and (16) into Equation (13),

$$[u_\phi^i \ u_\psi^i]^T = [1 \ R]^T \quad (23)$$

Since the modal mass m_1 is

$$m_1 = [u_\phi^i \ u_\psi^i] \begin{bmatrix} I_x & -I_{zx} \\ -I_{zx} & I_z \end{bmatrix} \begin{bmatrix} u_\phi^i \\ u_\psi^i \end{bmatrix} \quad (24)$$

Substituting Equations (16) and (23) into Equation (24),

$$m_1 = \frac{I_x I_z - I_{zx}^2}{I_z}$$

Moreover, substituting Equation (20) and $I_x/I_z = p$ into this,

$$m_1 = \frac{I_z}{p \sin^2 \alpha + \cos^2 \alpha} \quad (25)$$

Substituting Equations (23) and (25) into Equation (12), the steady-state frequency response turns out to be given by the following equation.

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \\ \phi \\ \psi \\ \delta \\ R \end{bmatrix} = \frac{(p \sin^2 \alpha + \cos^2 \alpha) F_\phi}{I_z (-\omega^2 + \omega_1^2)} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ R \end{bmatrix} \quad (26)$$

Substituting Equation (16) into the first equation of (18), the resonant frequency ω_1 is

$$\omega_1^2 = \frac{(K_{\phi\phi} K_{\psi\psi} - K_{\phi\psi}^2) I_z}{(I_x I_z - I_{zx}^2) K_{\psi\psi}}$$

Substituting Equations (20), (21), $I_x/I_z = p$ and $K_x^*/K_y^* = q$,

$$\omega_1^2 = \frac{K_x^* (p \sin^2 \alpha + \cos^2 \alpha)}{I_z (q \sin^2 \beta + \cos^2 \beta)} \quad (27)$$

Next denoting the angle between the axis of rotational vibration and x-axis by δ ,

$$\tan \delta = -R \quad (28)$$

Substituting Equation (28) into Equation (22),

$$\tan \delta = \frac{(1-p) \tan \alpha}{p \tan^2 \alpha + 1} \quad (29)$$

On the other hand, since

$$\tan (\alpha - \delta) = \frac{\tan \alpha - \tan \delta}{1 + \tan \alpha \tan \delta} \quad (30)$$

Substituting Equation (29) into Equation (30),

$$\tan (\alpha - \delta) = p \tan \alpha \quad (31)$$

Similarly,

$$\tan (\beta - \delta) = q \tan \beta \quad (32)$$

According to Equations (31) and (32), the positional relationship among the inclination of the principal axis of inertia α , that of the principal axis of elasticity β and that of rotation axis δ is shown in Table 1.

Summarizing the above description as the design procedure, since p and α are determined by the characteristics of engines as rigid body, q and β are selected according to Equation (22), and setting the resonant frequency according to Equation (27), K_x^* is determined. The inclination of the axis of rotational vibration δ at this time is given by Equation (29).

Table 1. Relationship between α , β , and δ

$\begin{pmatrix} \alpha \\ \beta \\ \delta \end{pmatrix} < \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$		$p < 1$		$1 \leq p$	
		$\beta < 0$	$0 < \beta$	$\alpha < 0$	$0 < \alpha$
$q < 1$	$\alpha < 0$	$\alpha < \beta < 0$ $\beta < \delta < 0$	--	--	$\beta < \delta < 0 < \alpha$
	$0 < \alpha$	--	$0 < \delta < \alpha$ $\beta < \delta < \alpha$	$\alpha < \beta < \delta$	--
$1 \leq q$	$\beta < 0$	--	$\beta < 0 < \delta < \alpha$	$\delta < 0$	--
	$0 < \beta$	$\alpha < \delta < 0 < \beta$	--	--	$\delta < 0$

EXAMPLE OF APPLICATION

In Figure 3, the example of a motorcycle to which the orthogonal engine mount system is applied is shown. The engine is a water-cooled, two stroke cycle, two cylinder type with a 180° crankshaft, with the cylinders inclined 24° forward of vertical. The unbalanced primary inertia moment is perpendicular to this, namely, around the axis inclining by 24° downward to the chassis front. Also, the principal axis of inertia of the engine inclines as largely as 30° upward to the chassis front.

As for the mounting of this engine, two cylindrical vibration-isolating rubbers each are arranged in front and rear locations of the engine almost symmetrically in the lateral direction with narrow distance. According to the theory described before, the principal axis of elasticity is inclined by 10° downward to the chassis front, and the rubbers are designed so as to work as compression members against the movement of the engine in the fore/aft and vertical directions.

By this method, the vibration due to the unbalanced primary inertia moment can be limited to rolling only, and setting this resonant rotating speed below the usable operating speed range, the supporting stiffness in other direction than rolling can be sufficiently secured simultaneously. Hereinafter, the course of its verification is described.

SIX-DEGREE OF FREEDOM SYSTEM MODEL—The system model comprises the matrix component for the engine, expressed as a rigid body, and the scalar components expressed with springs and hysteretic dampers. Of these, the moments and products of inertia which are matrix elements for the engine were determined by pendulum test, and the scalar components for mounts were obtained by curve fitting the frequency response functions with the sinusoidal excitation of 10 - 200 Hz using an electrohydraulic exciter.



Fig. 3 Motorcycle

In Figure 4, one example is shown. Next, the analysis of the system model was performed using the building block approach (hereafter, written as BBA).

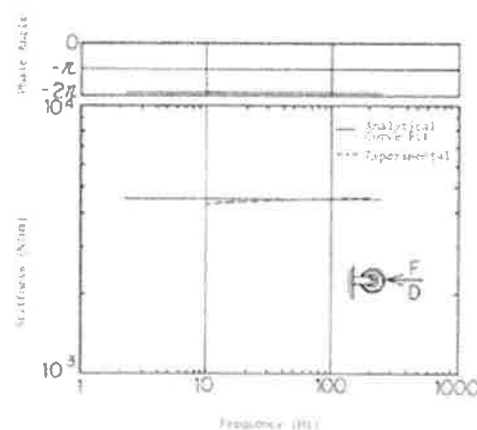


Fig. 4 Typical Engine Mount Frequency Response Function

In addition, as shown in Figure 5, the excitation test was carried out for the system comprising the engine connected to its rubber engine mounts and the engine mount connection points to the fixture constrained to the ground by applying sinusoidal excitation with an electromagnetic exciter and measuring the frequency response function at 10 - 200 Hz. Thereafter, the natural frequencies and natural vibration modes were determined by the modal analysis techniques.

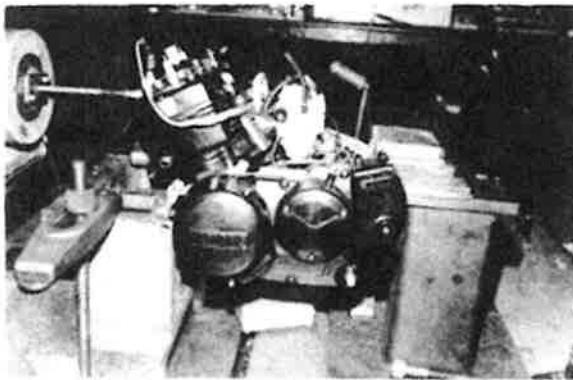


Fig. 5 Excitation Test for Engine on Mounts to Ground

Table 2 lists the comparison between the natural frequencies determined from the test data, and predicted natural frequencies calculated by the system model. Figure 6 and 7 show the comparison between test and calculated results for the two exciter location's driving point frequency response functions. Good correlation was obtained.

Table 2. Predicted and Measured Natural Frequencies for Engine on Mounts to Ground

Measured Freq. (Hz)	Predicted Freq. (Hz)	Percent Deviation	Mode Description
35.0	34.5	+1.4	Roll
50.5	47.0	+6.9	Lateral
76.5	73.8	+3.5	Yaw
100.0	107.5	+7.5	Fore/Aft
109.0	103.9	-4.7	Bounce
180.0	166.4	-7.6	Pitch

Next, since the excitation corresponding to the primary inertia moment and the measurement of rotational displacements of the center of gravity of the engine are difficult, the simulation using the system model mentioned before was performed for the steady-state frequency response. Figure 8 shows the results. As the engine mount system for a vehicle, this system has extraordinarily high supporting stiffness such as the highest natural frequency is 180 Hz. However, the resonant frequency is only one at the lowest natural frequency of 35 Hz, and it is known that the vibration-isolating function is well satisfied.

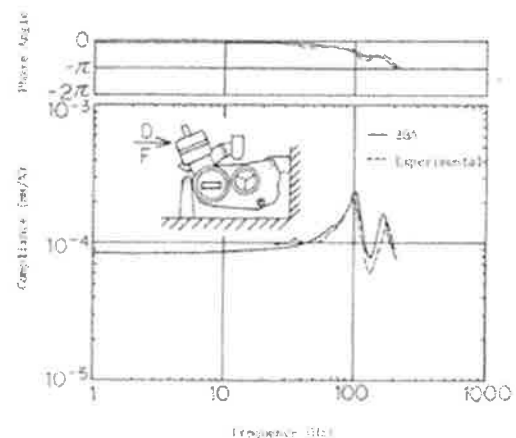


Fig. 6 Predicted and Measured Frequency Response Function for Engine on Mounts to Ground

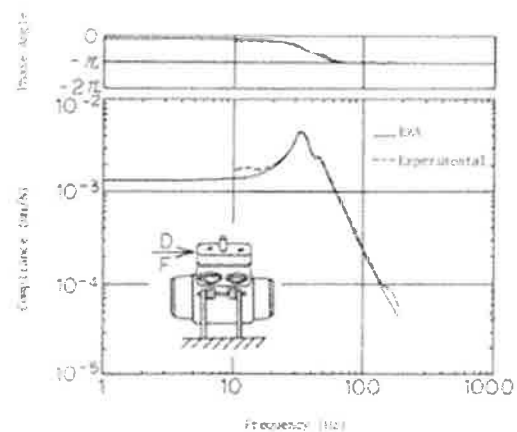


Fig. 7 Predicted and Measured Frequency Response Function for Engine on Mounts to Ground

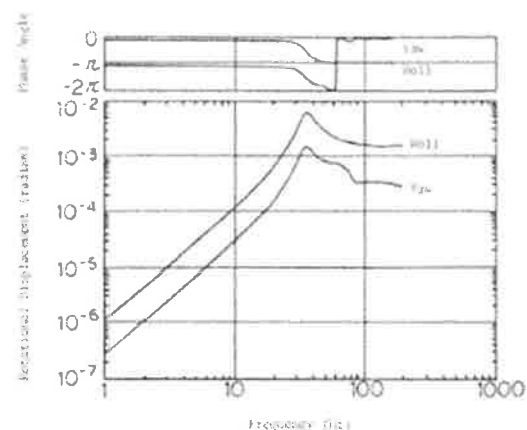


Fig. 8 Predicted Frequency Response for Rotational Displacement of Engine on Mounts to Ground

Chassis with 29 modes (0-500 Hz)
Muffler with 15 modes (0-1000 Hz)

MULTI-DEGREE OF FREEDOM SYSTEM MODEL-Now, of course, the chassis of motorcycles do not have infinite mass, and are not rigid bodies. Thereupon, as shown in Figure 9, the motorcycle was divided into a number of components, and respective components were represented by the mathematical models corresponding to their characteristics, thus the system model was built up again by BBA.

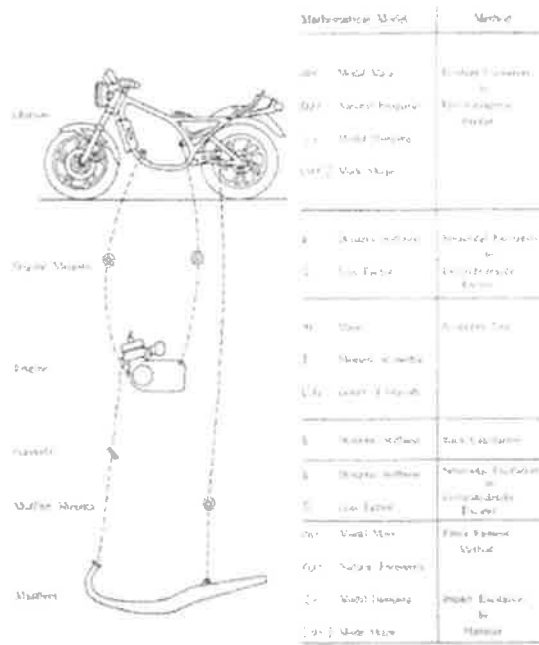


Fig. 9 Divided Components for Motorcycle System Model

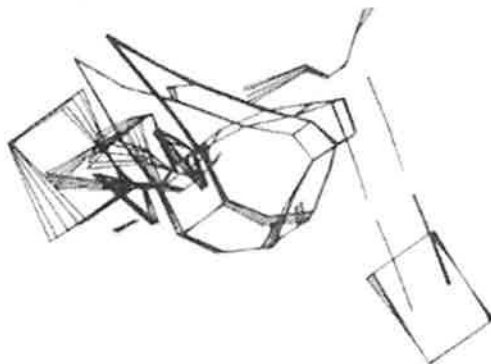


Fig. 10 Typical Animated Mode Shape of Bare Chassis

A chassis and mufflers are the modal components expressed with 29 modes (0 - 500 Hz) and 15 modes (0 - 1000 Hz) respectively. In Figure 10, an example of the mode animation for a bare chassis is shown. Muffler mounts and gaskets are scalar components, and engine mounts and the engine are the same as those used for the six-degree of freedom system model mentioned before.

Also, respective nodal points of the model are assumed to have three degrees of freedom, translation only, but the center of gravity of the engine, gaskets and the connection points of mufflers to the engine are expressed with six degrees of freedom including rotation. Among them, in the case of mufflers, it is difficult to determine the degree of freedom of rotation by the excitation test, therefore, the modeling was performed by the eigenvalue analysis using the Finite Element Method. Also for similar reasons, the stiffnesses of the gaskets were determined by back calculation from the comparison between the results of simulation and excitation test data for the muffler on gasket to the ground subsystem model.

On the other hand, the excitation test on the full vehicle was carried out by sinusoidal excitation in the range of 10 - 200 Hz using an electromagnetic exciter, similar to the case of the six-degree of freedom system model. In Figure 11, the comparison of a measured and calculated driving point frequency response functions for the lateral cylinder head excitation is shown, and relatively good correlation is seen.

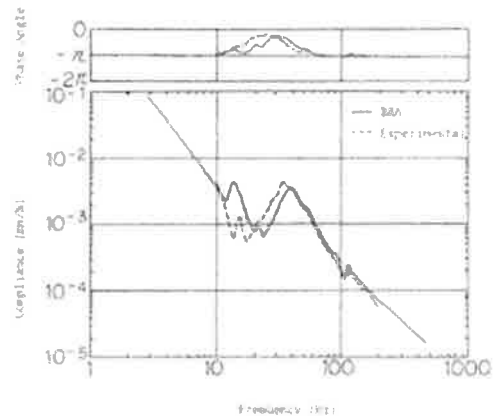


Fig. 11 Predicted and Measured Frequency Response Function for Full Vehicle

From the above description, it was found that the resonant frequency at 35 Hz in the six-degree of freedom system model was hardly shifted in the multi-degree of freedom system model. This seems to be because the moment of inertia of the chassis is sufficiently large compared with the moment of inertia of the engine.

Also, the resonance of the mufflers appeared at 108 Hz, but by positioning the muffler mount at the node of muffler vibration, the transmission of vibration to the chassis was suppressed. Figure 12 shows the vibration mode of mufflers.

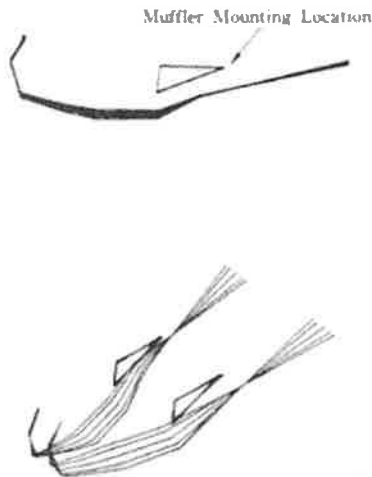


Fig. 12 Animated Mode Shape of Muffler (108 Hz)

MEASUREMENT OF VIBRATION OF ACTUAL VEHICLES
-In order to observe the effectiveness of the orthogonal engine mount system, vibration measurement was performed on a vehicle using this system and a vehicle using conventional engine mounts. First, the vibration evaluating method is explained, then the results of evaluation are reported.

Generally, the sense of human bodies to vibration does not coincide with measured vibration data. The reason for this and the method of solution are described in various literatures. Since 1974, Yamaha has established and used its own vibration measurement and analysis system based on the vibration greatness level (VGL) evaluation method (3) proposed by Miwa.

First, the vibration acceleration level (VAL) is defined by the following equation.

$$VAL_i = 20 \log (a_i/a_{ref}) \quad (33)$$

Where a_i : measured rms 1/3 octave band vibration acceleration (g)

a_{ref} : reference rms vibration acceleration, $10^{-3}g$

Selecting the VAL at 20 Hz defined by Equation (33) as the reference, the level at other frequencies which are sensorially equivalent to this VAL are defined by the quantity called "vibration greatness level", its unit is VGL.

This corresponds to loudness level, PHON in psychoacoustics. In Figure 13, the equal sensation curve for the random vibration on the hand adopted in this system is shown.

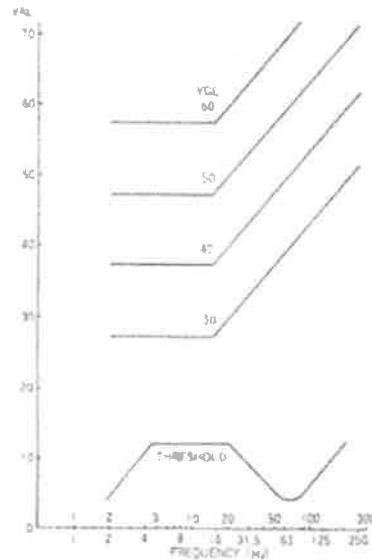


Fig. 13 Equal Sensation Curve for the Random Vibrations of 1 - Octave Band on the Hand

$$VGL = \begin{cases} VAL_i + 2 & \{i < 12\} \\ VAL_i - 2(i-13) & \{i \geq 12\} \end{cases} \quad (34)$$

Where i : 1/3 octave band No.

The VGL value corresponds to the interval scale in psychology, and the addition law does not hold. Accordingly, VGL must be converted to the scale which has true ratio property. This ratio scale is defined by the quantity called "vibration greatness", with VG as its units. This corresponds to the loudness in psychoacoustics, SONE. Assuming 1VG to be 40 VGL, the relation between VG and VGL is expressed by following two equations.

$$\log VG = \begin{cases} 0.03VGL - 1.2 & \text{(below 1 VG)} \\ 0.023VGL - 0.92 & \text{(above 1 VG)} \end{cases} \quad (35)$$

Next, usually vibration includes many frequency spectrums, and since the inhibition effect exists in the sense of human bodies, the VG value of respective frequencies are summed up by weighted addition. Namely,

$$VG_{total} = VG_{max} + 0.13 [(2VG_i) - VG_{max}] \quad (36)$$

VGL_{total} is determined substituting VG_{total} obtained by Equation (36) into Equations (35).

Incidental, the following techniques are used in this system for determining the "combined" effect of multi-axis vibration. This combined indicator can be determined from rms 1/3 octave band vibration acceleration in three direction, a_{xi} , a_{yi} , a_{zi} , simply by

$$a_i = \sqrt{a_{xi}^2 + a_{yi}^2 + a_{zi}^2} \quad (37)$$

To obtain practical VGL, a_i is calculated using Equation (37), then Equations (33), (34), (35) and (36) are applied successively, and finally Equation (35) is applied.

Now, the results of the evaluation using this system at the three positions where human bodies are in contact with motorcycles, the handle bars, the seat and the foot rests (handle bars and foot rests are represented by the left side only) are shown in Figure 14. It is known that the reduction of vibration has been achieved except at the engine speed corresponding to 35 Hz, near 2000 rpm, at which the result was worse than a vehicle with bolt-connected engine. Also, the effect was more conspicuous at higher engine speed.

SUMMARY

Utilizing the dynamic characteristics of a rigid body as an engine and the static characteristics of springs as mounts, a new method of engine vibration isolation was devised. Its effectiveness was confirmed both theoretically and experimentally. Also, the application to actual motorcycle was analyzed using modal analysis and BBA techniques, and very satisfactory results were obtained. Moreover, it should be added that in this motorcycle, low vibration allows the chassis to be light weight, and together with the high performance of the engine, it has obtained a high reputation in Japan and Europe.

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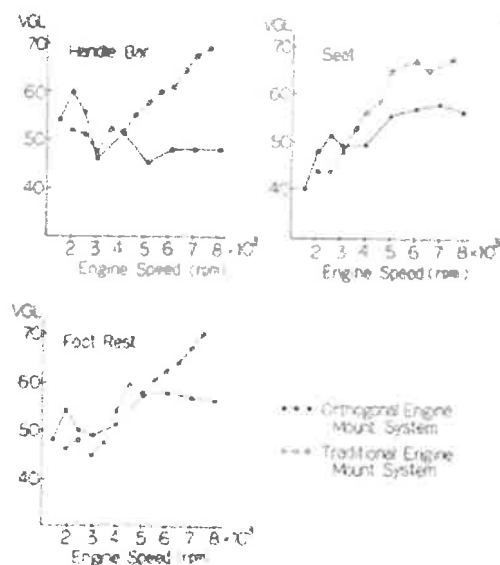


Fig. 14 -Result of VGL Evaluation for Actual Motorcycle

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